

Roll No.

TEC-801

**B. TECH. (EC) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

COMPUTER ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain basic structure of computer and their functional unit with neat and clean block diagram. (CO1)
(b) What is a CPU ? Write its major component and explain *three* types of reference instructions. (CO1)
(c) Explain any *four* arithmetical instructions and any *two* addressing modes. (CO1)
2. (a) Convert the following hexadecimal number in octal, binary and decimal number : (CO1, CO6)
 - (i) 1234H
 - (ii) ABCDH

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- (b) Write multiplication using Booth algorithm. (CO1, CO6)
- (c) Write a short note on ALU design. (CO1, CO6)
3. (a) What is meant by a hardwired implementation of a control unit ?
Explain briefly. (CO4)
- (b) Write a short note on microprogrammed control unit. (CO4)
- (c) Define the following terms : (CO4)
- (i) Interpretation
- (ii) Control unit
4. (a) Explain memory mapping. Draw a mapping of 64 KB of memory by
8 KBs of memory. (CO3, CO5)
- (b) What is the need of cache memory ? Discuss its advantages.
(CO3, CO5)
- (c) What is an associative memory ? How can associative memory be
realized ? (CO3, CO5)
5. (a) What do you mean by I/O mapped I/O and memory mapped I/O ? What
are their advantages and disadvantages ? (CO2, CO4)
- (b) What is DMA ? Explain DMA mode of data transfer using a suitable
block diagram. (CO2, CO4)
- (c) What short notes on the following : (CO2, CO4)
- (i) Parallel processing
- (ii) Pipelining

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TPE-014

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023

TESTING FOR VLSI CIRCUITS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss various trends that plays a major role to affect testing. (CO1)

(b) Explain the following terms : (CO1)

(i) Behavioural Fault

(ii) Bridging Fault

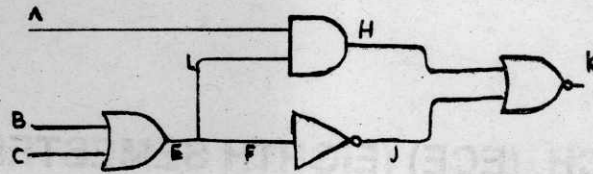
(iii) Cross-point Fault

(iv) Path-Delay Fault

(c) Explain the modeling of signal states with the help of an example.(CO1)

P. T. O.

2. (a) Analyze the combinational SCOAP for the circuit given below : (CO2)

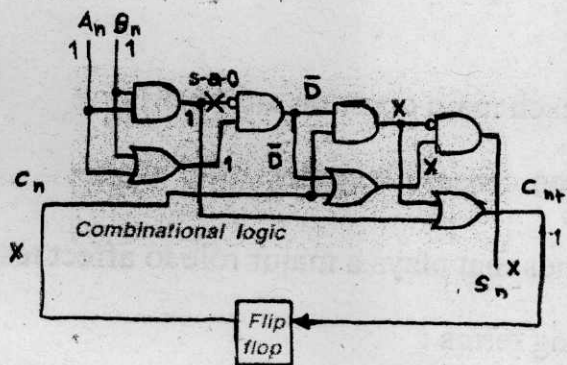


- (b) Explain Redundancy Identification (RID) with the help of an example.

(CO2)

- (c) Determine a test sequence for the s-a-0 fault on the output line of the flip-flop in the circuit shown below :

(CO2)



3. (a) Discuss memory density and defect trends. (CO3)
- (b) Write a short note on I_{DDQ} test and discuss its limitations. (CO3)
- (c) Discuss the various types of coupling faults in detail. (CO3)

(3)

4. (a) Explain scan architecture and technique. Also discuss its disadvantages. (CO4, 6)
(b) Differentiate between Analog testing and Digital testing. (CO4, CO6)
(c) Discuss various methods used for analog testing considering present day scenario. (CO4, CO6)
5. (a) Describe BIST architecture and its benefits. (CO1, CO5)
(b) Discuss various modeling levels and types of simulators. (CO1, CO5)
(c) Explain the statistical methods for fault simulation. (CO1, CO5)

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TPE-015

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023

WIRELESS SENSOR NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the *ten* differences between wireless sensor networks and cellular networks. (CO1)

(b) What are Wireless Sensor Networks (WSNs) ? Explain its properties. How is WSN special class of wireless ad hoc networks ? (CO1)

(c) Explain the basic operating principle of wireless ad hoc networks. How is ad hoc networks useful in collaborative computing and emergency applications ? (CO1)

P. T. O.

2. (a) What are the unique challenges of Wireless Sensor Networks (WSN) ?
Explain briefly. (CO2, CO3)
- (b) Explain the following applications of WSNS : (CO2, CO3)
- (i) Industrial Applications
 - (ii) Logistics Applications
 - (iii) Volcano activity monitoring
 - (iv) Urban Applications
- (c) Explain the features, architecture and devices used in Zigbee Protocols.
(CO2, CO3)
3. (a) Explain the "Superframe Structure" of the IEEE 802.15.4 MAC protocol. (CO3, CO4)
- (b) Explain the B²-MAC (Berkeley MAC) protocol in detail. (CO3, CO4)
- (c) Explain the following : (CO3, CO4)
- (i) PEGASIS (Power Efficient Gathering for Sensor Information Systems)
 - (ii) Binary Scheme
4. (a) What is the "Data Dissemination" process in WSN ? Explain the 'Rumor Routing'. (CO4)
- (b) Differentiate between "Flooding" and "Gossiping" data dissemination process. (CO4)

(3)

- (c) How does the "Data Gathering" process take place in WSN ? Explain the "Chain-Based Three Level Scheme" with for data gathering process.

(CO4)

5. (a) What is the design principle of WSNs ? What are the main hardware components are used for single node architecture of WSNs ?

(CO5)

- (b) What is the role of Gateways in WSN ?

(CO5)

- (c) How and why is Internet to WSNs communication process advantageous ?

(CO5)

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TPE-016

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023

CMOS ANALOG CIRCUIT DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain in detail about large signal and small signal models of MOS amplifier. (CO1)
- (b) Derive the equation of depletion width and built-in potential of $p-n$ junction diode. (CO1)
- (c) Derive the equation of I-V characteristics of NMOS device. (CO1)
2. (a) Derive the voltage gain of common source stage with source degeneration using small and large signal analysis. (CO2)

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- (b) Explain the working of common source stage with resistive load and derive the equation of voltage gain. (CO2)
- (c) Explain the working of source follower amplifier. (CO2)
- 3. (a) Explain the working of cascode current source and cascode amplifier. (CO3)
- (b) Explain in detail about large signal and small signal analysis of differential amplifier. (CO3)
- (c) Explain the construction and working of one stage op-amp. (CO3)
- 4. (a) Explain the working of cascode current mirror in detail. (CO4)
- (b) Explain the working of Wilson current mirror in detail. (CO4)
- (c) Explain the working of Widlar current source in detail. (CO4)
- 5. (a) Derive the expression of closed loop gain, input impedance and output impedance of voltage-voltage feedback topologies. (CO5/CO6)
- (b) Explain in detail about the working of PLL circuit. (CO5/CO6)
- (c) Derive the expression of closed loop gain, input impedance and output impedance of voltage-current feedback topologies. (CO5/CO6)

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TPE-008

B. TECH. (E and C) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
NEURAL NETWORK AND MACHINE LEARNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the concept of Artificial Neural Network and its use for pattern recognition. (CO1)
- (b) Explain the different types of learning methodologies for Neural Network. (CO1)
- (c) Explain the use of linear regression. (CO1)
2. (a) With example, explain autoassociative neural network. (CO2)
- (b) With example, explain heteroassociative neural network. (CO2)
- (c) Explain the method fast learning method for neural network. (CO2)

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(2)

3. (a) With example, explain the use of Radial Basis Function Neural Network for classification. (CO3)
- (b) With examples, explain Relevance Vector Machine. (CO3)
- (c) With examples, explain the use of Support Vector Machine for classification. (CO3)

4. (a) Explain the MAXNET architecture and illustrate its use for pattern :

$$X = [x_1, x_2, x_3, x_4] = [0.1, 0.3, 0.8, 0.6]$$

- (b) Explain the use of Learning Vector Quantization Neural Network for four bit inputs and two clusters : (CO4)

$S_1 = [1, 0, 0, 0]$ belongs to cluster C_1

$S_2 = [0, 1, 0, 0]$ belongs to cluster C_1

$S_3 = [1, 0, 0, 0]$ belongs to cluster C_1

$S_4 = [0, 0, 0, 1]$ belongs to cluster C_2

$S_5 = [0, 0, 1, 1]$ belongs to cluster C_2

- (c) With an example, explain use of Self-Organising Map for classification.
5. (a) Explain Hopfield Neural Network with an suitable example. (CO5)
- (b) Explain the Boltzmann machine architecture. (CO5)
- (c) Explain the architecture of recurrent neural network. (CO5)